W4U

W O R K F O R Y O U

Product Specification

Comprehensive Product, Design & Technical Specification

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INTRODUCTION

Purpose

This document is the comprehensive product specification for W4U (Work For You), a peer-to-peer platform enabling micro-entrepreneurs to verify their identity and skills, build trusted reputations, and receive payments. It serves as the single source of truth for product, design, and engineering teams.

Product Vision

"Every person has value to offer. We exist to build trust and ensure people are recognised and rewarded for what they contribute."

Product Definition

"W4U verifies identity and skills, builds trusted reputations, and enables payments — empowering micro-entrepreneurship as a means to building rewarding livelihoods."

Scope

This specification covers the MVP (Minimum Viable Product) and V2 releases. The MVP focuses on identity verification, basic profiles, ratings/reviews, QR-based tipping, and wallet functionality. V2 adds the full marketplace, AI-powered matching, multilingual support, and community recognition features.

Document Conventions

- MVP — features required for initial launch (Months 1–9).
- V2 — features for second release (Months 9–18).
- Future — planned but not yet specified in detail.
- [P] = Provider-facing, [C] = Consumer/Community-facing, [A] = Admin-facing.

USER PERSONAS

Persona 1: Sipho — The Informal Tradesperson

Age: 34
Location: Johannesburg, Gauteng
Occupation: Painter (14 years experience, no formal employment)
Digital literacy: Moderate — uses WhatsApp, Facebook, some SnapScan
Language: isiZulu (primary), English (conversational)

Context

Sipho stands outside Builders Warehouse every morning hoping for work. He's an excellent painter but has no way to prove it to strangers. He earns R300–800 per day when he gets work, but many days he gets nothing.

Goals

- Prove his skills and identity to potential customers.
- Build a reputation that carries from job to job.
- Get found by people looking for a painter without standing on a street corner.
- Receive payments digitally (safer than carrying cash).

Frustrations

- Starts from zero trust with every new customer.
- Losing work to less skilled people who happen to be in the right place.
- No way to show his portfolio of completed work.
- Cash-only means he can't serve customers who don't carry cash.

Persona 2: Lindiwe — The Domestic Worker

Age: 42
Location: Midrand, Gauteng
Occupation: Nanny and cleaner (works for 3 families)
Digital literacy: Low-moderate — WhatsApp only, limited English
Language: Sesotho (primary), basic English

Context

Lindiwe has been doing domestic work for 15 years. Her current employers found her through word-of-mouth. She's reliable and trusted but has no formal references she can show to new potential employers.

Goals

- Have a verified profile that proves her track record.

- Get additional work to fill open days.
- Receive tips and bonuses digitally.
- Build a record that helps her negotiate better pay.

Persona 3: Thabo — The Car Guard

Age: 26
Location: Sandton, Gauteng
Occupation: Car guard at a shopping centre
Digital literacy: High — smartphone user, social media, mobile banking
Language: English, isiZulu

Context

Thabo guards cars in a busy shopping centre parking lot. He's friendly, diligent, and helps shoppers with bags. Many regulars appreciate him, but new visitors don't know if he's trustworthy.

Goals

- Make it easy for people to tip him digitally.
- Build visible proof that he's a great car guard.
- Stand out from less reliable guards in the area.

Persona 4: Sarah — The Homeowner / Consumer

Age: 38
Location: Rosebank, Johannesburg
Occupation: Marketing manager
Digital literacy: High
Language: English

Context

Sarah needs a reliable painter for her house, a part-time gardener, and occasionally tips car guards and petrol attendants. She currently relies on WhatsApp group recommendations and takes a chance on whoever shows up.

Goals

- Find verified, reviewed service providers near her.
- Know that the person she's hiring is who they say they are.
- Leave feedback to help others (and reward good work).
- Pay digitally — easy and trackable.

Persona 5: David — The Community Champion

Age: 55
Location: Centurion, Pretoria
Occupation: Retired, active in his residential estate committee
Digital literacy: Moderate
Language: Afrikaans, English

Context

David notices people in his community who do good work — the man who voluntarily sweeps the street corner, the gardener who goes above and beyond. He wants to recognise and reward them, and make sure the wrong people don't claim credit.

Goals

- Vouch for people he knows and trusts in the community.
- Tip community value creators easily.
- Help verify that the right person gets credit for the right work.

USER JOURNEYS

Journey 1: Provider Onboarding (Sipho)

1. Downloads W4U from Google Play.
2. Selects language (isiZulu). App switches to isiZulu.
3. Enters mobile number. Receives OTP. Verifies.
4. Chooses "I offer services" (vs. "I'm looking for services").
5. AI-guided ID verification: takes photo of SA ID book, then selfie. AI matches face and extracts details.
6. Trust level advances to "ID Verified".
7. AI-assisted profile creation: Sipho speaks about his skills in isiZulu. AI transcribes and generates a bilingual profile (isiZulu + English).
8. Uploads 5 photos of completed painting work.
9. Optionally uploads trade certificate for "Skills Verified" status.
10. Generates personal QR code. Prints it to display at Builders.
11. Sets service area, availability, and indicative pricing.
12. Profile goes live. Sipho is discoverable in searches.

Journey 2: Consumer Finds & Hires (Sarah)

13. Opens W4U app. Searches "painter near Rosebank".
14. Results show verified providers sorted by rating, proximity, and trust level.
15. Taps Sipho's profile. Sees: ID Verified badge, 4.7-star rating, 23 reviews, 8 community vouches, portfolio photos.
16. Reads reviews from other homeowners.
17. Taps "Contact" — sends message via in-app chat.
18. Agrees on scope, price (R2,500), and date.
19. After job completion, Sarah pays R2,500 via W4U wallet.
20. Leaves a 5-star review with comments on quality, punctuality, and cleanliness.
21. Sipho receives payment (minus 2% platform fee) in his W4U wallet.

Journey 3: QR Tipping (Thabo the Car Guard)

22. Thabo displays his personal W4U QR code on a lanyard around his neck.

23. A shopper finishes loading groceries. Thabo helped carry bags.
24. Shopper scans Thabo's QR code with their phone camera.
25. Opens W4U (or web fallback for non-app users). Sees Thabo's mini-profile: name, photo, 4.5 stars.
26. Selects tip amount (R10) and confirms.
27. Optionally leaves a quick star rating (takes 2 seconds).
28. Thabo receives instant notification: "You received a R10 tip!"

Journey 4: Community Vouching (David)

29. David opens W4U and navigates to his community's neighbourhood page.
30. Sees a list of providers active in his area.
31. Finds Blessing, who cleans the street corner every morning.
32. Taps "Vouch" — confirms he personally knows Blessing and has seen her work.
33. Writes a short vouch statement: "Blessing has been cleaning our corner for 2 years. Reliable and hardworking."
34. Blessing's profile now shows "Vouched by 12 community members".
35. David tips Blessing R50 from the community tab.

FEATURE SPECIFICATION

4.1 Onboarding & Registration [P][C]

Release: MVP

The onboarding flow must be completable in under 5 minutes for basic registration, and under 10 minutes including ID verification and profile creation.

Requirements

- Mobile number registration with OTP verification.
- Language selection at first launch (11 SA languages + English default).
- User type selection: “I offer services” / “I’m looking for services” / “Both”.
- Progressive disclosure: don’t ask for everything upfront. Core identity first, then encourage profile completion over time.
- Skip-able steps with persistent nudges to complete profile.

Acceptance Criteria

- User can register and reach the home screen in < 3 taps after OTP.
- Language preference persists across sessions.
- Incomplete profiles show a completion percentage bar.

4.2 Identity & Verification Engine [P]

Release: MVP (basic), V2 (advanced)

4.2.1 ID Verification (MVP)

- User takes a photo of their SA ID document (smart ID card, ID book, or driver’s licence).
- AI extracts: full name, ID number, date of birth, photo.
- User takes a selfie.
- AI performs facial match (selfie vs. ID photo). Minimum confidence threshold: 85%.
- ID number validated against check digit algorithm (Luhn mod 10 for SA IDs).
- On success: user advances to “ID Verified” trust tier.
- On failure: user is prompted to retry or request manual review.

4.2.2 Skills Verification (MVP)

- User uploads photos/PDFs of certificates, diplomas, or trade qualifications.
- AI performs OCR to extract institution name, qualification, date, holder name.
- Cross-reference holder name against verified ID.

- V2: Automated verification against issuing institution databases where APIs exist.
- On success: user advances to "Skills Verified" trust tier.

4.2.3 Community Vouching (V2)

- Any verified user can "vouch" for another user they know personally.
- Vouch includes: relationship type, duration known, free-text statement.
- Minimum 5 vouches from unique verified users = "Community Vouched" tier.
- Anti-gaming: vouchers must be ID-verified. Mutual vouching (A vouches B and B vouches A) counts as only 1 vouch.

4.2.4 Trust Tiers

Tier	Requirements	Badge	Visibility
Unverified	Account created, mobile verified	Grey outline	Limited: can browse but not transact
ID Verified	SA ID verified + selfie match	Green shield + checkmark	Full: can transact, receive tips, appear in search
Skills Verified	ID Verified + at least 1 qualification verified	Green shield + star	Priority listing in skill-based searches
Community Vouched	ID Verified + 5 unique community vouches	Green shield + people icon	Featured in neighbourhood feeds
Fully Vetted	All of the above + background check	Gold shield	Top of search, premium trust badge

4.3 Profile & Portfolio [P]

Release: MVP

Profile Fields

Field	Type	Required	Notes
Display name	Text	Yes	From ID verification or user-entered
Profile photo	Image	Yes	From selfie verification or upload
Bio / About	Text (500 chars)	No	AI-assisted: voice-to-text in home language
Skills / services offered	Multi-select + custom	Yes (providers)	From predefined categories + free text
Service area	Map radius	Yes (providers)	Default 10km, adjustable 1–50km
Availability	Weekly schedule	No	Day-of-week + time blocks
Indicative pricing	Range or hourly	No	Optional, shown in search results
Portfolio photos	Up to 20 images	No	Before/after encouraged

Qualifications	Document uploads	No	Linked to verification engine
Languages spoken	Multi-select	No	From 11 SA languages
Trust tier	Auto-calculated	N/A	Based on verification status
Rating	Auto-calculated	N/A	Average of all reviews
Reviews	List	N/A	Linked from feedback system
Community vouches	Count + list	N/A	Linked from vouching system
QR code	Auto-generated	N/A	Unique per user, links to profile

AI-Assisted Profile Creation (V2)

- Provider opens profile builder and taps microphone icon.
- Speaks in their home language describing their skills and experience.
- AI transcribes, translates to English, and generates a professional bio.
- Provider reviews and edits (in either language).
- Profile is stored in original language + English. Displayed to consumers in their preferred language.

4.4 Reputation & Feedback System [P][C]

Release: MVP (ratings/reviews), V2 (vouching, AI analysis)

4.4.1 Ratings

- 1–5 star overall rating after each completed transaction or interaction.
- Dimension ratings (optional): Punctuality, Quality of Work, Communication, Value for Money.
- Quick-rate mode: single star tap (for tipping interactions). Full-rate mode: stars + dimensions + written review (for service engagements).
- Ratings below 3 stars require a written explanation (minimum 20 characters).

4.4.2 Reviews

- Written reviews attached to ratings. Visible on provider profile.
- Reviews are immutable once posted (no editing). Reviewer can add a follow-up comment.
- Provider can respond publicly to any review (once).
- AI flags: fake review detection, abusive language filter, sentiment analysis.

4.4.3 Trust Score Algorithm (V2)

Each provider's trust score is a weighted composite:

Signal	Weight	Description

Verification tier	25%	Higher tiers = higher base score
Average rating	25%	Weighted by recency (recent reviews count more)
Review volume	15%	More reviews = more confidence in the score
Community vouches	15%	Quality and quantity of vouches
Transaction history	10%	Volume and consistency of completed transactions
Response rate	5%	How quickly and reliably they respond to enquiries
Profile completeness	5%	Fully completed profiles score higher

4.5 Digital Wallet & Payments [P][C]

Release: MVP

4.5.1 Wallet

- Every user gets a W4U wallet upon registration.
- Wallet balance displayed on home screen.
- Fund wallet: bank transfer (EFT), instant EFT, card payment.
- Withdraw to bank account: standard (1–2 business days) or instant (small fee).
- Transaction history with filters: date, type, amount, counterparty.
- Exportable statements (PDF) for financial record-keeping.

4.5.2 Payment Types

Type	Description	Fee	Release
Tip	One-off micro-payment, typically via QR scan	1.5%	MVP
Service payment	Agreed-upon payment for completed work	2.5%	MVP
Wallet-to-wallet	Direct transfer between W4U users	1.5%	MVP
Escrow payment	Held until service is confirmed complete	2.5%	V2
Recurring payment	Scheduled repeating payments (e.g., weekly gardener)	2.0%	V2
Community fund	Pooled neighbourhood fund for shared value creators	1.5%	V2

4.5.3 QR Code System

- Each provider has a unique, permanent QR code.
- QR code links to: w4u.co.za/p/{username} (web-accessible even without the app).
- Scanning opens: mini-profile + tip/pay interface.
- Non-app users: web fallback with card payment option.
- QR code is downloadable, printable, and shareable.

- Physical QR cards: W4U will produce branded lanyards / table cards for providers (partnership with print vendors).

4.6 Service Marketplace [P][C]

Release: V2

4.6.1 Search & Discovery

- Search by: keyword, service category, location, rating, verification level, availability, price range.
- Location-based results with map view and list view.
- Sort by: relevance (AI-scored), rating, distance, price.
- Filter tags: "ID Verified", "Skills Verified", "Community Vouched", "Available Now".
- AI-powered search: natural language queries ("I need someone to paint my lounge next Saturday") parsed into structured search.

4.6.2 Service Categories (Initial)

Category	Sub-categories
Home & Property	Painting, Plumbing, Electrical, Handyman, Tiling, Roofing, Fencing
Domestic	Cleaning, Nanny/Childcare, Cooking, Laundry/Ironing, Elder Care
Garden & Outdoor	Gardening, Landscaping, Pool Maintenance, Tree Felling
Vehicle	Car Guarding, Car Wash, Minor Repairs, Panel Beating
Community	Street Cleaning, Recycling, Security, Neighbourhood Watch
Personal	Hairdressing, Tailoring, Tutoring, Pet Care, Errands
Skilled Trades	Welding, Carpentry, Bricklaying, Plastering
Hospitality	Petrol Attendant, Waiter, Barista, Event Staff

4.6.3 Job Posting

- Consumers can post a job request (description, location, budget, timeframe).
- Matching providers receive notifications.
- Providers can express interest and send a quote.
- Consumer selects a provider and confirms.
- Post-completion: both parties rate each other.

4.6.4 Messaging

- In-app messaging between matched/interested parties.
- Real-time with push notifications.
- AI-powered translation (e.g., consumer writes in English, provider sees isiZulu).

- Message templates for common interactions ("I'm interested", "When are you available?", "Job complete").
- No contact info sharing before mutual agreement (anti-spam, safety).

4.7 Community Recognition [C]

Release: V2

- Neighbourhood feeds showing activity, top providers, and recent vouches.
- Leaderboards: top-rated providers by category in each neighbourhood.
- Achievement badges: "50 Five-Star Reviews", "Trusted by 25+ People", "1 Year on W4U".
- Community funds: residents pool tips for shared value creators (e.g., the street cleaner). Distributed weekly.
- Community attribution: if someone claims to be providing a community service (e.g., cleaning), community members can confirm or dispute.

4.8 Admin & Moderation [A]

Release: MVP (basic), V2 (full)

- Dashboard: user growth, transaction volume, verification queue, reported issues.
- Manual verification review queue for edge cases (AI confidence < 85%).
- Content moderation: review flagged reviews, profiles, and messages.
- User management: suspend, warn, ban users.
- Fraud investigation tools: view transaction patterns, network graphs, anomaly flags.
- Financial reporting: revenue, fees collected, wallet balances, withdrawal activity.
- Analytics: funnel metrics, retention, NPS, category performance.

SCREEN-BY-SCREEN DESCRIPTIONS

Detailed wireframe descriptions for each key screen. Visual wireframes are provided in the companion file (w4u-wireframes.jsx).

S1: Splash / Language Selection

Purpose: First-time launch experience

- W4U logo centred, animation: W and U slide in from sides, 4 drops from above.
- Language selector: grid of 11 SA language buttons + English.
- Single CTA: "Get Started".
- Background: soft green gradient.

S2: Registration

Purpose: Account creation

- Mobile number input with SA country code (+27) pre-filled.
- OTP entry screen (6 digits, auto-verify from SMS).
- User type selector: three illustrated cards ("I Offer Services", "I Need Services", "Both").

S3: ID Verification

Purpose: Identity verification flow

- Camera viewfinder with overlay showing ID card placement guide.
- Instruction text in selected language.
- After photo: AI processing animation (shield icon filling up).
- Selfie screen: face outline guide, instruction to match ID photo pose.
- Success screen: green checkmark, "ID Verified!" badge animation, trust tier indicator advancing.
- Failure screen: friendly retry prompt with tips ("Make sure your ID is well-lit").

S4: Profile Builder

Purpose: Create service provider profile

- Top: profile completion percentage bar (green fill).
- Sections as expandable cards: About Me, Skills, Service Area, Availability, Photos, Pricing.
- Microphone button on "About Me" for voice-to-text (V2).
- Photo upload: grid layout, tap-to-add, drag-to-reorder.
- Service area: interactive map with draggable radius circle.
- Save button (sticky bottom bar) + "Preview My Profile" link.

S5: Home Screen (Provider)

Purpose: Provider dashboard

- Top bar: W4U logo, notification bell (with badge count), profile avatar.
- Wallet summary card: balance, last transaction, “Withdraw” button.
- My QR Code card: tap to enlarge/share/download.
- Recent activity feed: latest tips received, reviews, enquiries.
- Quick stats: total earnings (week/month), rating, profile views.
- Bottom nav: Home, Search, QR Scan, Messages, Profile.

S6: Home Screen (Consumer)

Purpose: Consumer dashboard

- Top bar: W4U logo, notification bell, profile avatar.
- Search bar (prominent): “What do you need help with?”
- Category quick-links: icon grid (Painting, Cleaning, Gardening, Car Guard, etc.).
- Nearby top-rated providers: horizontal scroll cards.
- Recent interactions: providers you’ve tipped or hired.
- Bottom nav: Home, Search, QR Scan, Messages, Profile.

S7: Search Results

Purpose: Browse service providers

- Toggle: List view / Map view.
- Filter bar (horizontal scroll): Verified, Rating 4+, Available Now, Price Range.
- Provider cards: photo, name, trust badge, rating (stars + count), distance, price range, primary skill.
- Map view: pins at provider locations, tap to see mini-card.

S8: Provider Profile (Public View)

Purpose: Full profile visible to consumers

- Hero: large profile photo, name, trust badge, star rating.
- Trust section: verification badges with explanations.
- About section: bio text (in consumer’s language).
- Portfolio: photo gallery (tap to expand).
- Reviews: sorted by recency, with star breakdown chart.
- Community vouches: count + expandable list of vouchers.
- Action buttons (sticky bottom): “Contact”, “Tip”, “Hire”.

S9: QR Scan / Tip

Purpose: Scan QR to tip or pay

- Camera viewfinder (full-screen) with QR overlay guide.
- On scan: provider mini-card slides up (photo, name, rating).
- Tip amount selector: preset buttons (R5, R10, R20, R50, Custom).
- Quick-rate: 1–5 stars (optional, single tap).
- Confirm button. Success animation (gold coin dropping into wallet).

S10: Wallet

Purpose: Financial management

- Balance: large number, prominent.
- Action buttons: Fund Wallet, Withdraw, Send.
- Transaction history: scrollable list with icons (tip received, payment sent, withdrawal, funding).
- Each transaction: counterparty name/photo, amount, date, type.
- Tap transaction for details (receipt view).
- Export button (PDF statement).

NON-FUNCTIONAL REQUIREMENTS

Performance

- App launch to interactive: < 3 seconds on mid-range Android devices.
- Search results: < 2 seconds for location-based queries.
- ID verification processing: < 15 seconds end-to-end.
- Payment processing: < 5 seconds.
- QR scan to tip confirmation: < 10 seconds total flow.

Scalability

- Architecture must support 1 million concurrent users.
- Horizontal scaling for all services.
- Database sharding strategy for multi-country expansion.

Availability

- 99.9% uptime target (< 8.7 hours downtime per year).
- Graceful degradation during load shedding (offline queuing for non-critical actions).

Security

- End-to-end encryption for messages and payment data.
- PCI DSS compliance for payment handling (via certified payment gateway).
- Biometric data (facial recognition) encrypted at rest and in transit. Not stored permanently — used for verification match only, then discarded.
- POPIA compliance: explicit consent for data collection, right to erasure, data minimisation.
- Rate limiting and DDoS protection on all APIs.
- Two-factor authentication for wallet withdrawals above threshold.

Accessibility

- WCAG 2.1 AA compliance target.
- Voice-to-text and text-to-voice for low-literacy users (V2).
- High contrast mode.
- Large text support.
- Screen reader compatibility.

Localisation

- 11 official SA languages supported (UI strings, AI translation, voice input).
- Right-to-left text support not required (no SA languages are RTL).
- Currency: South African Rand (ZAR) as default. Multi-currency support for pan-African expansion.
- Date/time format: South African conventions (DD/MM/YYYY, 24-hour).

Data & Analytics

- Event tracking on all user interactions (anonymised).
- Funnel analytics: registration > verification > profile complete > first transaction.
- Real-time dashboard for operational metrics.
- Data warehouse for long-term analytics and ML model training.

TECHNICAL ARCHITECTURE (APPENDIX)

This appendix provides the technical architecture specification sufficient for an engineering team to begin system design and implementation planning.

System Architecture Overview

W4U follows a microservices architecture deployed on cloud infrastructure (AWS or Azure). Each core domain is an independent service with its own database, communicating via an API gateway and event bus.

Service Map

Service	Responsibility	Database	Key Integrations
Auth Service	Registration, OTP, sessions, tokens	PostgreSQL	Twilio (SMS), Firebase Auth
Identity Service	ID verification, facial match, document OCR	PostgreSQL + S3 (temp images)	AWS Rekognition, Tesseract OCR, Home Affairs API
Profile Service	User profiles, portfolios, skills	PostgreSQL + Elasticsearch	Identity Service, Storage Service
Reputation Service	Ratings, reviews, vouches, trust scores	PostgreSQL + Redis (cache)	AI Service (fraud detection)
Payment Service	Wallets, transactions, QR codes, statements	PostgreSQL (ACID)	Peach Payments / Stitch, bank APIs
Marketplace Service	Search, matching, job posting, categories	Elasticsearch + PostgreSQL	Profile Service, AI Service
Messaging Service	In-app chat, push notifications	MongoDB + Redis (pub/sub)	Firebase Cloud Messaging, AI Service (translation)
AI Service	ML models, NLP, fraud detection, translation	Model store (S3) + feature store	All services (inference API)
Admin Service	Dashboard, moderation, analytics	Read replicas + analytics DB	All services (admin API)
Notification Service	Push, SMS, email notifications	Redis queue	Firebase, Twilio, SendGrid

Infrastructure

- Cloud: AWS (primary) with multi-AZ deployment in af-south-1 (Cape Town region).
- Container orchestration: Kubernetes (EKS) for all services.
- API Gateway: Kong or AWS API Gateway with rate limiting, auth, and routing.
- Event bus: Apache Kafka for inter-service communication and event sourcing.
- CDN: CloudFront for static assets and QR code pages.
- CI/CD: GitHub Actions with automated testing, staging, and production deployment.
- Monitoring: Datadog or Grafana + Prometheus. PagerDuty for alerts.

Data Models

Core entity relationship overview:

User

Field	Type	Notes
id	UUID	Primary key
phone_number	String	Unique, E.164 format
user_type	Enum	PROVIDER, CONSUMER, BOTH
display_name	String	From ID or user-entered
language_preference	String	ISO 639-1 code
trust_tier	Enum	UNVERIFIED, ID_VERIFIED, SKILLS_VERIFIED, COMMUNITY_VOUCHED, FULLY_VETTED
trust_score	Float	0.0–5.0, computed
average_rating	Float	0.0–5.0, computed
total_reviews	Integer	Count
total_vouches	Integer	Count
wallet_id	UUID	FK to Wallet
qr_code_url	String	Unique URL slug
created_at	Timestamp	
verified_at	Timestamp	Nullable
is_active	Boolean	Soft delete flag

Verification

Field	Type	Notes
id	UUID	Primary key
user_id	UUID	FK to User
type	Enum	ID_DOCUMENT, SELFIE_MATCH, QUALIFICATION, BACKGROUND_CHECK
status	Enum	PENDING, PROCESSING, VERIFIED, FAILED, MANUAL_REVIEW
confidence_score	Float	AI confidence (0.0–1.0)
document_type	String	e.g., SA_ID_CARD, DRIVERS_LICENCE, TRADE_CERT
extracted_data	JSONB	OCR results, structured
reviewer_id	UUID	Nullable, for manual reviews
created_at	Timestamp	
verified_at	Timestamp	Nullable

Review

Field	Type	Notes
id	UUID	Primary key
reviewer_id	UUID	FK to User (who left the review)
provider_id	UUID	FK to User (who is being reviewed)
transaction_id	UUID	FK to Transaction (nullable for tips)
overall_rating	Integer	1–5
punctuality	Integer	1–5, nullable
quality	Integer	1–5, nullable
communication	Integer	1–5, nullable
value_for_money	Integer	1–5, nullable
comment	Text	Required if rating < 3
provider_response	Text	Nullable, one response allowed
is_flagged	Boolean	AI or user flagged
created_at	Timestamp	

Wallet & Transaction

Field	Type	Notes
wallet.id	UUID	Primary key
wallet.user_id	UUID	FK to User
wallet.balance	Decimal(12,2)	Current balance in ZAR
wallet.status	Enum	ACTIVE, FROZEN, CLOSED
transaction.id	UUID	Primary key
transaction.from_wallet_id	UUID	FK to Wallet (nullable for funding)
transaction.to_wallet_id	UUID	FK to Wallet (nullable for withdrawal)
transaction.type	Enum	TIP, SERVICE_PAYMENT, TRANSFER, FUND, WITHDRAW, ESCROW
transaction.amount	Decimal(12,2)	Gross amount
transaction.fee	Decimal(12,2)	Platform fee
transaction.net_amount	Decimal(12,2)	Amount received by payee
transaction.status	Enum	PENDING, COMPLETED, FAILED, REFUNDED
transaction.reference	String	Human-readable ref
transaction.created_at	Timestamp	

API Design

RESTful API with JSON payloads. Authentication via JWT tokens. API versioning: /api/v1/...

Key Endpoints

Method	Endpoint	Description	Auth
POST	/api/v1/auth/register	Register with phone + OTP	None
POST	/api/v1/auth/verify-otp	Verify OTP and get JWT	None
POST	/api/v1/verification/id	Submit ID document for verification	JWT
POST	/api/v1/verification/selfie	Submit selfie for facial match	JWT
POST	/api/v1/verification/qualification	Submit qualification document	JWT
GET	/api/v1/profile/{id}	Get user profile (public view)	Optional
PUT	/api/v1/profile/me	Update own profile	JWT
POST	/api/v1/profile/me/photos	Upload portfolio photo	JWT
GET	/api/v1/search	Search providers (query, location, filters)	Optional
POST	/api/v1/reviews	Create a review	JWT
GET	/api/v1/reviews/provider/{id}	Get reviews for a provider	Optional
POST	/api/v1/vouches	Vouch for a user	JWT
GET	/api/v1/wallet/me	Get wallet balance and summary	JWT
POST	/api/v1/wallet/fund	Fund wallet (initiate payment)	JWT
POST	/api/v1/wallet/withdraw	Withdraw to bank account	JWT + 2FA
POST	/api/v1/transactions/tip	Send a tip	JWT
POST	/api/v1/transactions/pay	Service payment	JWT
GET	/api/v1/transactions/me	Transaction history	JWT
GET	/api/v1/qr/{username}	QR code landing page data	None
POST	/api/v1/messages	Send a message	JWT
GET	/api/v1/messages/conversations	List conversations	JWT

AI/ML Pipeline

Model Inventory

Model	Purpose	Input	Output	Infrastructure
ID Doc Classifier	Detect document type and authenticity	Camera image	Document type + authenticity score	TensorFlow Serving
OCR Extractor	Extract text fields from ID documents	Cropped document image	Structured JSON (name, ID#, DOB)	Tesseract + custom post-processor
Facial Matcher	Compare selfie to ID photo	Two face images	Match confidence (0–1)	AWS Rekognition or custom model
Review Fraud Detector	Identify fake or coordinated reviews	Review text + metadata	Fraud probability score	scikit-learn / XGBoost

Trust Score Engine	Compute dynamic trust scores	Multi-signal feature vector	Trust score (0–5)	Custom Python service
Translation Engine	Translate between 11 SA languages	Text + source/target lang	Translated text	Fine-tuned multilingual model
Smart Matcher	Rank providers for a given query	Query + provider features	Ranked list with scores	Learning-to-rank model
Fraud Transaction Monitor	Flag suspicious payment patterns	Transaction stream	Alert flags	Real-time streaming (Kafka + ML)

Security Architecture

- Authentication: JWT with short-lived access tokens (15 min) and long-lived refresh tokens (30 days). Refresh token rotation.
- Authorisation: Role-based access control (RBAC) — roles: user, provider, admin, super_admin.
- Data encryption: AES-256 at rest, TLS 1.3 in transit.
- PII handling: personally identifiable information stored in encrypted, access-controlled database columns. Audit logging on all PII access.
- Biometric data: facial images used for verification match only, then deleted within 24 hours. Not stored in user profiles.
- Payment security: PCI DSS compliance delegated to certified payment gateway. W4U never stores card numbers.
- POPIA compliance: consent management, data portability exports, right-to-erasure workflow.
- Penetration testing: quarterly third-party pen tests. Bug bounty programme from V2.

Release Plan

Release	Timeline	Key Features
MVP	Months 1–9	Registration, ID verification, basic profiles, ratings/reviews, QR tipping, wallet, basic search, admin dashboard
V2	Months 9–18	Marketplace (search, categories, job posting), AI matching, multilingual NLP, community vouching, trust scores, escrow payments, messaging with translation
V3	Months 18–30	Community recognition (leaderboards, badges, funds), voice profiles, advanced AI (demand prediction, safety scoring), recurring payments
V4	Months 30–42	Financial services (micro-loans, insurance referrals), multi-country support, advanced analytics, public API for third-party integration

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End of Document